

Prasanth PM

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Developer focused on Generative AI, Deep Learning, and full-stack application development. Skilled in building AI-powered systems, working with LLMs, embeddings, and model workflows. Interested in integrating machine learning into scalable backend applications and real-world products.

EDUCATION

Amrita Vishwa Vidyapeetham

Coimbatore, India

Bachelor of Technology - Computer Science and Engineering — GPA: 7.02

2023 - Present

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks

IIT Madras

Chennai, India

Bachelor of Science - Data Science

2023 - Present

- Relevant Coursework: Deep Learning and Gen AI, Bayesian statistics, Machine Learning

SRV Innovative Public School

Tamil Nadu

High School Diploma — Class XII (CBSE) – 91.8%

PROJECTS

Optimized Rainbow Table Generator Using B-Tree

Data Structures and Algorithms / Security

- Developed a customizable rainbow table generator for password hash reversal using SHA-1/MD5, enhanced with a B-tree-based GomuhryTree for faster hash lookup and collision management.
- Integrated configuration-driven charset handling, structured logging, and efficient chain reduction algorithms for scalable password-cracking experiments.

Academic Management Web Portal — User Interface Development

Frontend Developer (HTML, CSS, JavaScript)

- Developed responsive frontend web portal with role-based access for students and teachers using HTML, CSS, and JavaScript.
- Implemented features including course management, attendance tracking, assignment uploads, and grade viewing.

Smart Attendance System

Embedded C, STM32, RFID, SPI, I2C, DMA

- Engineered a high-performance attendance tracking system using 32-bit microcontroller with innovative bit manipulation technique, where each integer represents 32 students using DMA for efficient memory operations.
- Integrated RFID module via SPI and LED display via I2C protocols, designing a scalable system that handles real-time attendance marking with minimal memory footprint through efficient bit-level data representation.

QoE-Aware Multi-Tier Edge Offloading for Real-Time Visual SLAM and Object Detection

Python, YOLOv5, SLAM Libraries, OpenCV, SimPy, NumPy, Matplotlib

- Designed and simulated a dynamic multi-tier Edge Computing architecture (Tiers 1, 2, 3) to optimize the performance of real-time computer vision applications.
- Implemented a computational offloading strategy for critical tasks, specifically integrating YOLOv5 (Object Detection) and Visual SLAM (Simultaneous Localization and Mapping).
- Modeled the system using the SimPy discrete-event simulation library to analyze deployment trade-offs, focusing on minimizing latency and maximizing Quality of Experience (QoE).

TECHNICAL SKILLS

Programming languages: C/C++, Python, Java, Haskell, JavaScript, TypeScript.

Frameworks & Libraries: ReactJS, Express, Tailwind CSS, PyTorch.

Embedded Systems: STM32 Microcontrollers, RFID, SPI, I2C, DMA, GPIO, Embedded C

Databases: MySQL, MongoDB, Redis, Prisma, PostgreSQL.

Interests: Machine Learning, NLP, Blockchain, Algorithms.